GROUNDVALUE

AI property valuation for lending

Hero Housing Fintech Challenge
Problem Statement 1



The property valuation workflow

01

Collect property evidence

Transaction records, listings, title context and local infrastructure arrive in different formats.

02

Reconcile differences

Area definitions, duplicate listings and stale observations can distort comparisons.

03

Assess collateral risk

A lender needs a value range, its evidence and a clear reason to refer a case.

GroundValue brings the research and review into one property file.

A workspace for every property

Research

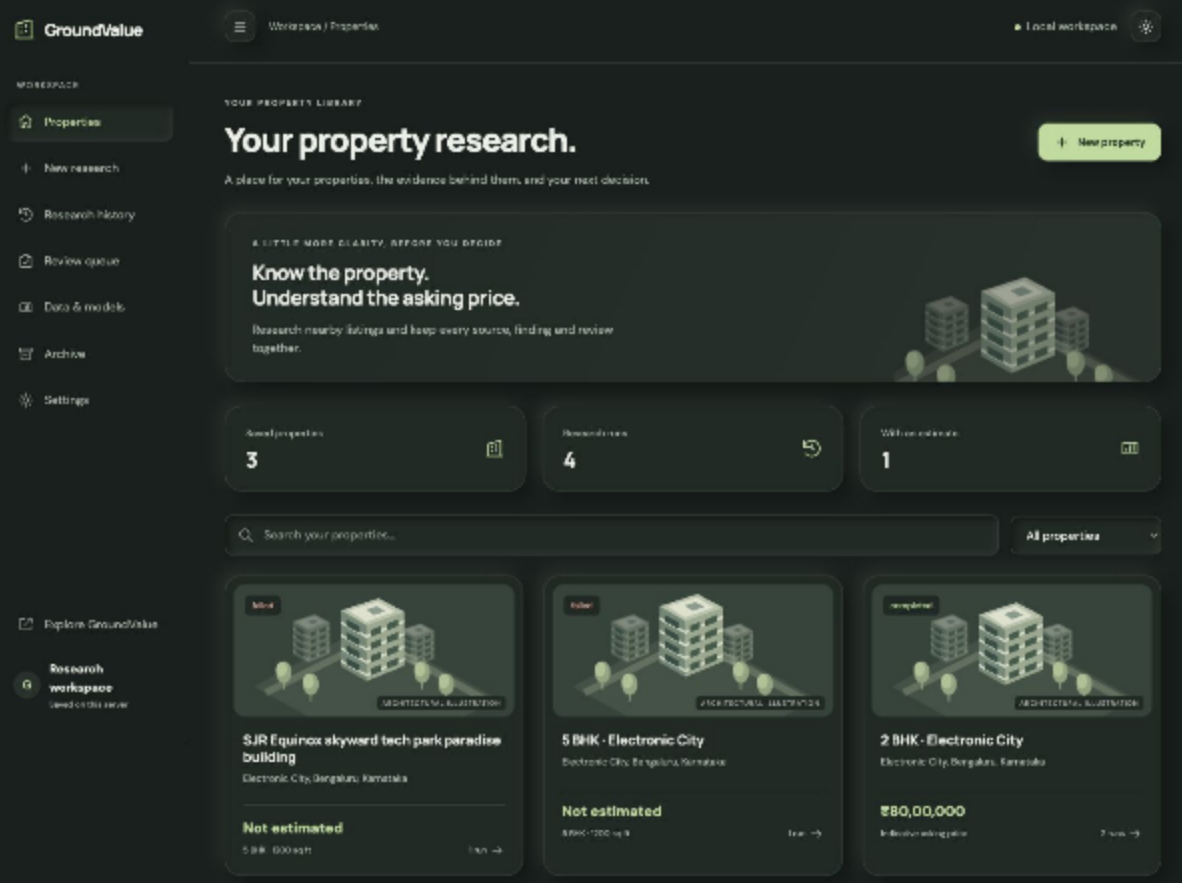
Create a property file and run source research.

Review

Open evidence, estimates and previous versions.

Retain

Keep the report and the reasoning together.



India coverage, by city and property type

01

Bengaluru

Residential apartments as the first validation cohort.

02

Hyderabad, Pune, NCR

Expand city models after local transaction and coverage checks.

03

Wider India

Extend through regional data partnerships and city-specific validation.

Residential

Area, age, configuration and neighbourhood

Commercial

Office and retail, with rents, occupancy and yields

Data collection and property linkage

01

Source evidence

Registered transactions
Government property records
Listings and market trends
Geospatial and satellite context

02

Clean and match

Standardise units and dates
Separate carpet and built-up area
Deduplicate buildings and listings
Link parcel, address and coordinates

03

Versioned property file

Source URL and observation date
Access basis and extraction record
Quality flags and missing fields
Features linked to the original evidence

Unstructured documents and pages feed the same schema as structured records.

Property details guide the research

One property identity

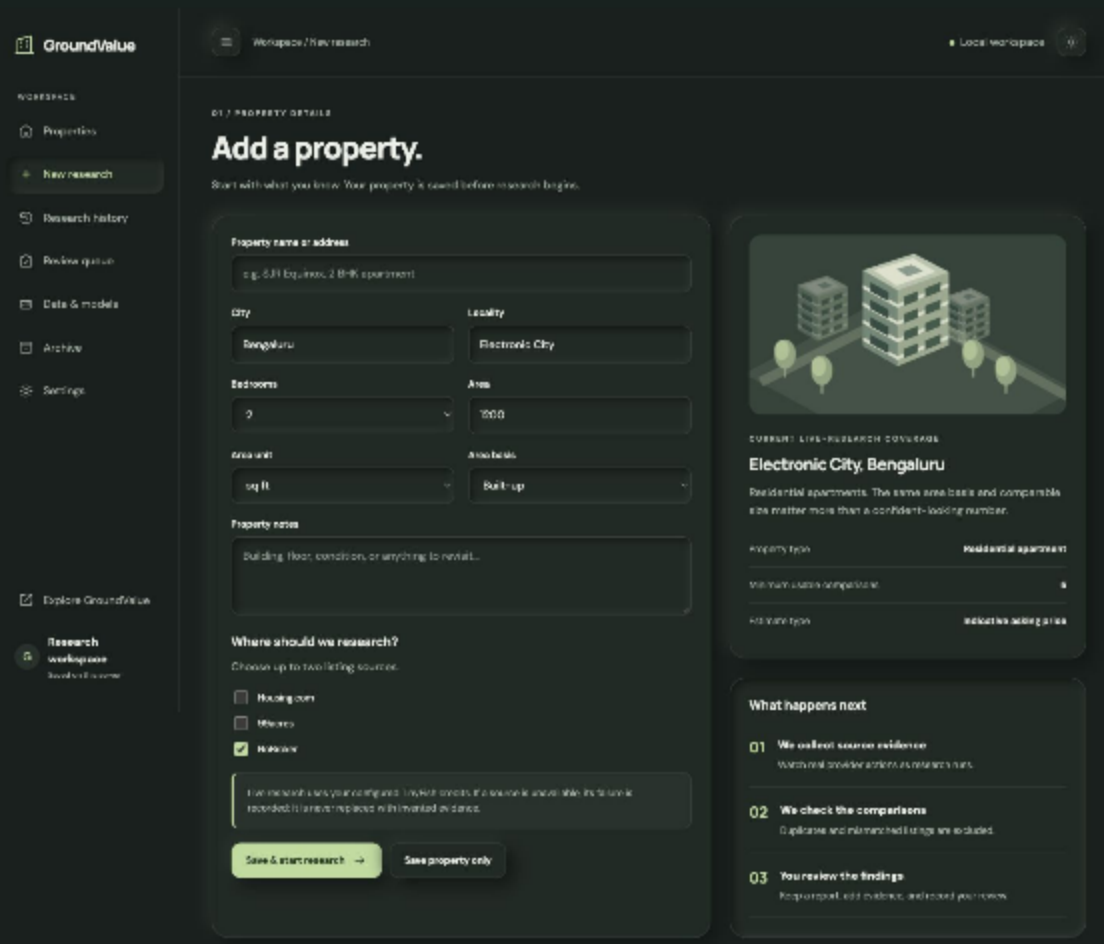
Location, configuration, area basis and property characteristics.

Source selection

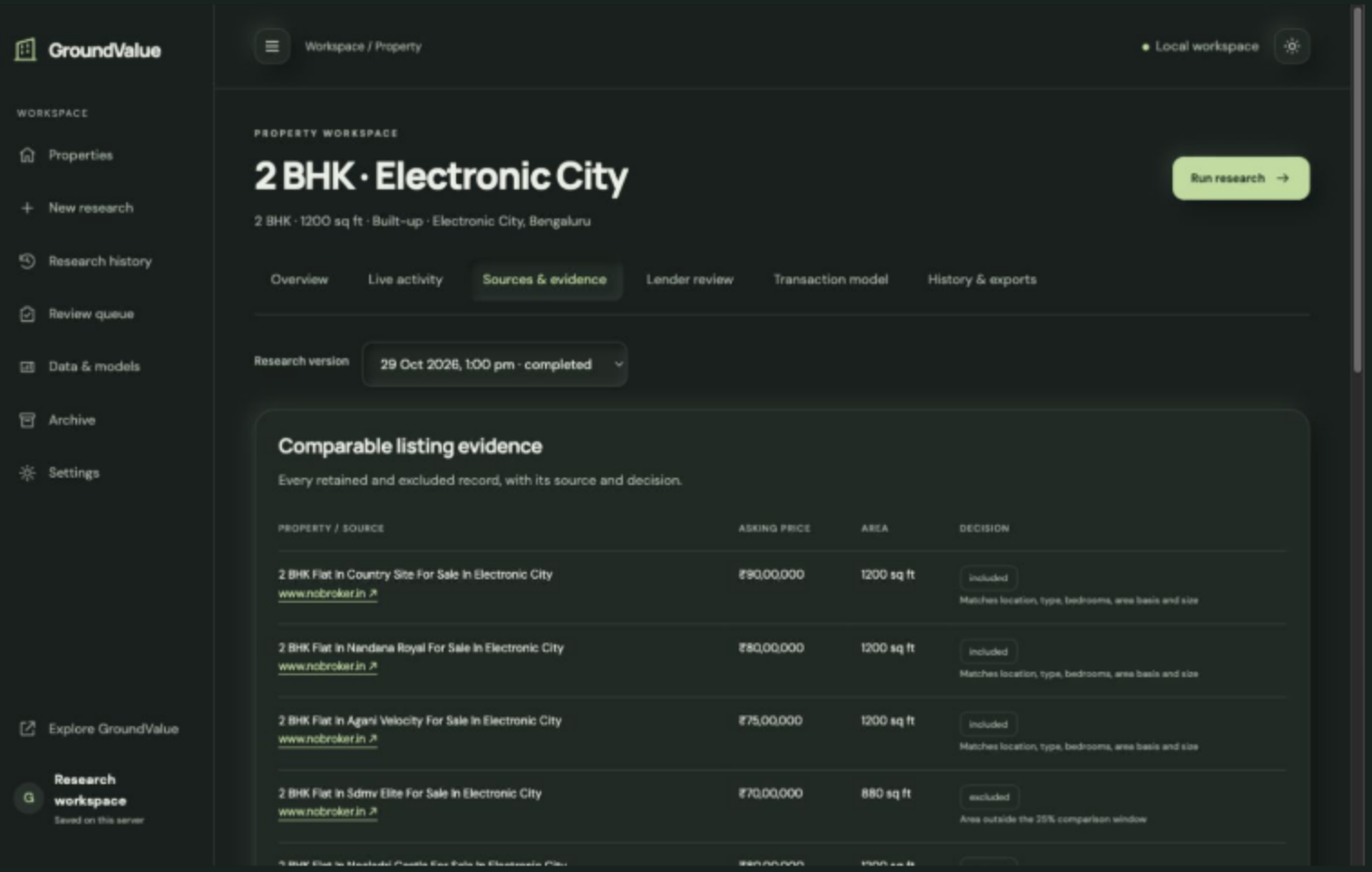
Research follows the selected sources and keeps the original inputs.

Reproducible runs

A new run creates a new version of the property research.



Comparable evidence stays inspectable



Match quality

Location, property type, size and area basis.

Exclusions

Every rejected comparison keeps its reason.

Sources

Open the source behind each observation.

The valuation model

INPUT

Property features

Area, bedrooms, age, coordinates, metro distance, city, property type and area basis.

LEARN

Transaction price

Log-price ridge regression provides an interpretable baseline. Gradient boosting is the challenger design.

OUTPUT

Value and uncertainty

A point estimate, calibrated interval and feature contributions, with an explicit coverage check.

Comparable market rates provide an independent benchmark for the model.

Accuracy measurement and model release

60%

Train

Fit preprocessing and model parameters.

20%

Calibrate

Estimate the residual interval.

20%

Test

Measure on untouched later records.

Time-ordered splits with separate buildings

MAE and RMSE in rupees. Median percentage error. R^2. Interval coverage.
Compare the same test records against the median-rate baseline and external AVMs where available.

A value estimate with its reasoning

Value range

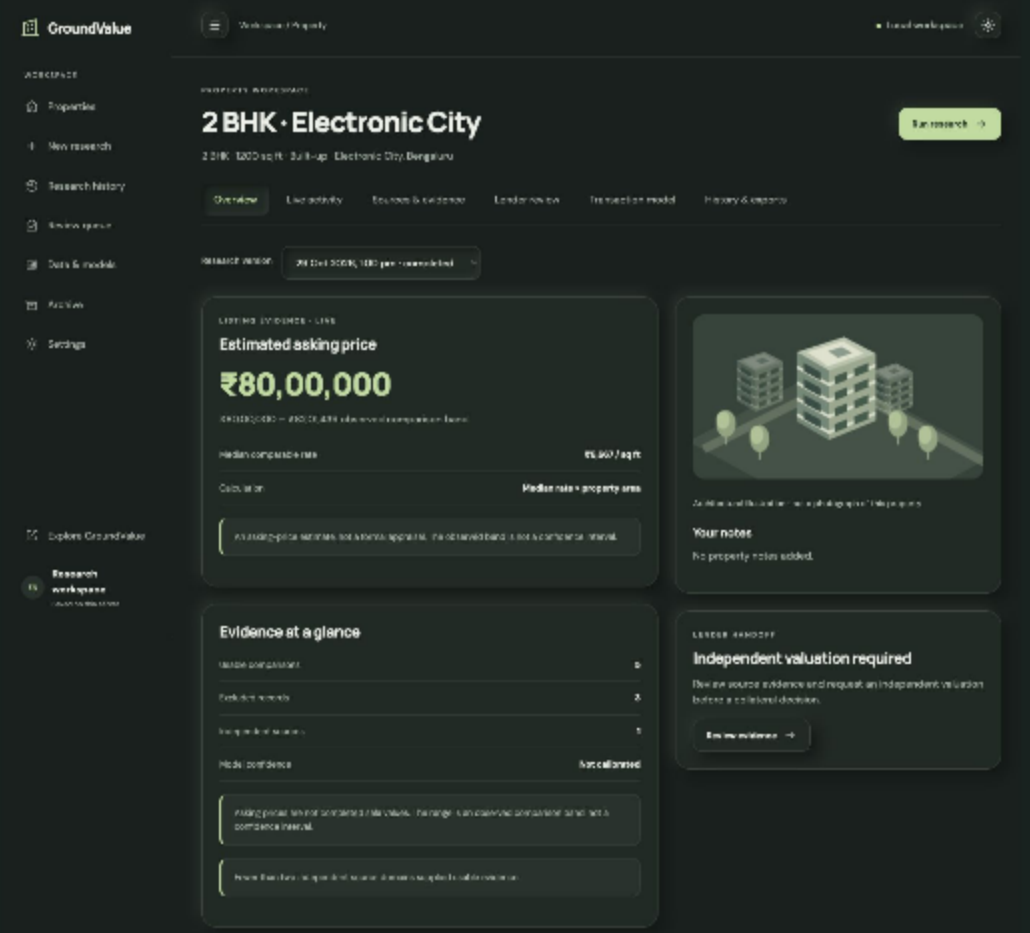
Show the estimate and the comparison band.

Evidence behind it

Retain selected comparables, source counts and calculation method.

Clear distinctions

Separate asking-price research from a transaction-trained valuation.



Lender review and escalation

Review the property file

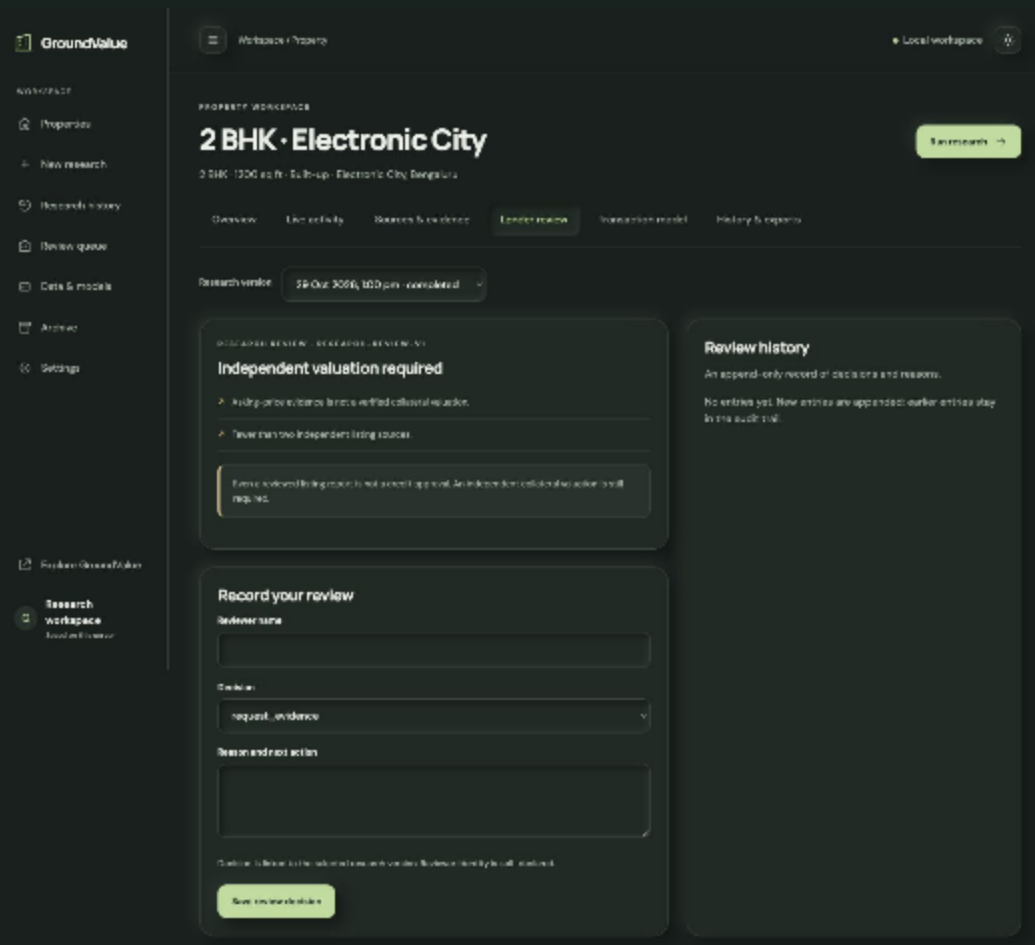
Check source quality, uncertainty and missing evidence.

Choose the next action

Request evidence, refer an independent valuer or record the research review.

Keep the decision trail

Attach the reviewer’s reason to the exact research version.



Automation and deployment architecture

Browser workspace

Property intake and review

Application API

Validation and case state

Research service

Source collection jobs

Model service

Prediction and checks

Persistent research store

Property versions, evidence, job state and review history survive browser sessions.

Operational controls

Track failed jobs, source freshness, model versions and regional coverage. Route exceptions to review.

Reports and audit history

Report handoff

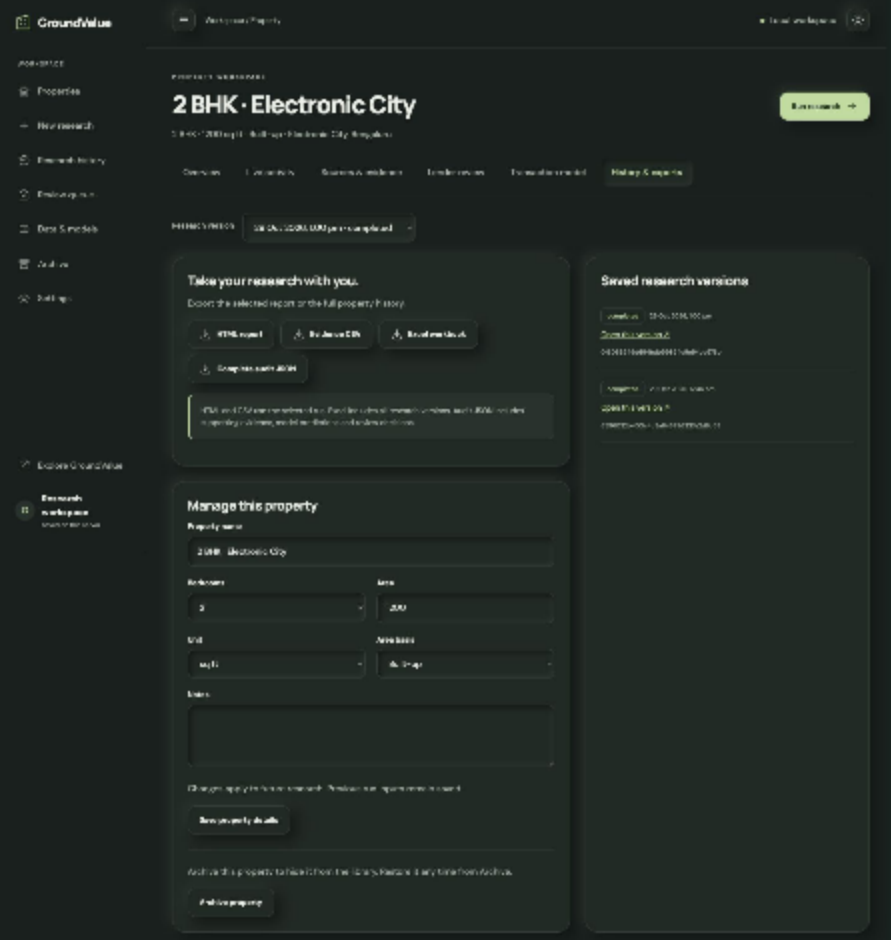
HTML report, evidence CSV and Excel workbook.

Complete audit record

Export property history, evidence, predictions and reviews as JSON.

Version integrity

Property edits apply to future research while earlier run inputs remain saved.



Pilot with a housing finance partner

01

Validate on lender data

Use authorised transactions and independent valuations across property segments.

02

Measure the benefit

Compare valuation error, research turnaround, manual effort and exception rates.

03

Expand by evidence

Broaden city and commercial coverage after accuracy and audit acceptance.

Integration: a property case in the loan workflow, with an API response and reviewable report.

GroundValue for digital property valuation

Research the property.
Explain the estimate.
Keep the evidence.

Solution coverage

India rollout and property scope
Structured and unstructured data
ML training and evaluation
Explainability and confidence
Automation and exception review
Lender workflow integration

Live demo

groundvalue.vercel.app

Source code

github.com/BENi-Aditya/GroundValue

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